Bayes Theorem: A Geometric Interpretation

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Introduction:

This document aims to provide a geometric, or visual, interpretation of Bayes' Theorem. It endeavors to guide readers from the fundamental principles to an intuitive understanding of the theorem. Starting with known probabilities, the paper derives Bayes' Rule using conditional probabilities and advances to a pragmatic application of Bayes' Theorem, addressing uncertain parameters represented as random variables and described by probability distributions. In doing so, I strive to bridge analytical descriptions with geometric interpretations, thus offering a comprehensive insight into the theorem's intrinsic dynamics. Ultimately, the goal is to lay down a theoretical yet intuitive foundation in Bayesian statistics.

Bayes Rule

Shown below, Bayes' Rule can be derived using known conditional probabilities. A geometric interpretation of the underlying mechanics is also provided in this section.

Conditional Probabilities:

Let A and B be events represented as subsets of sample space S, where S is the set of all possible outcomes. Set theoretically, let $A \subseteq S$ and $B \subseteq S$.

The conditional probability of event A, given event B is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \text{ where } P(B) > 0 \tag{1}$$

This relationship can be expressed visually as follows:

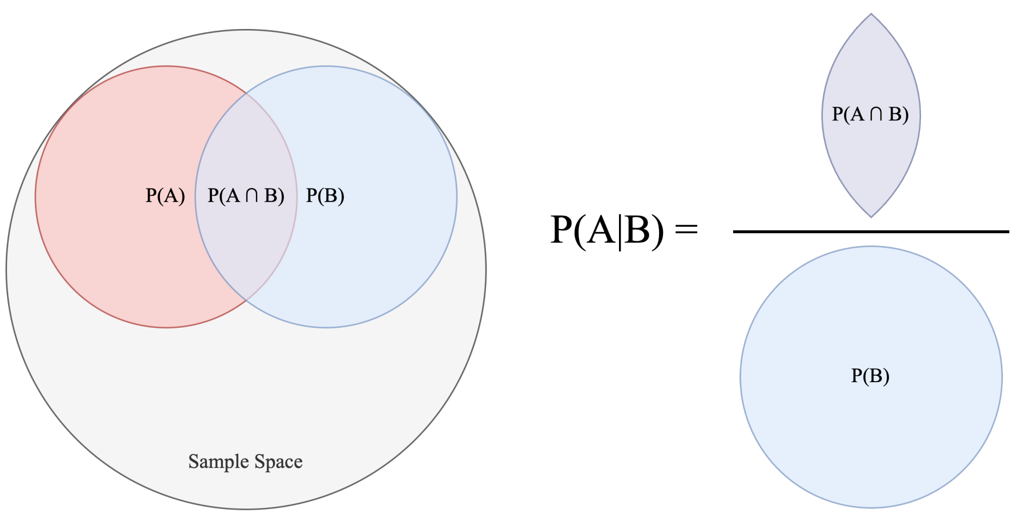

Figure 1: A visual representation of conditional probability

Shown above, some of the shapes are subsets of other shapes. Each shape represents an event, or set of outcomes, that an outcome can be categorized as. Each shape's area is used to represent the probability of an outcome being categorized into the event the shape represents. The sample space S represents all possible outcomes relevant to a specific, given context. Because S contains every possible outcome, an outcome must lie within S. Therefore, $P(S) = 1$. As a useful representation, the probability of an outcome being categorized as a particular event is a number that describes the certainty of that outcome taking on the specific properties that define that event. When it comes to conditional probabilities, the following question provides insight into their nature: "given you know B happened, what's the probability A also happened?" The added knowledge of B's occurrence confines the set of outcomes to those in B. Succinctly, $P(A \mid B)$ is the proportion of the probability mass in B that also lies in A. This proportion can be 0, 1, or anywhere in between. For example, if events A and B are mutually exclusive, then the probability that A occurs given a known B occurrence would be 0. If instead, A always occurs when B occurs, it would be 1.

Deriving Bayes' Rule:

Given the definition of conditional probability,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \text{ where } P(B) > 0 \tag{2}$$

Similarly,

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)}, \text{ where } P(A) > 0 \tag{3}$$

By equivalence,

$$P(A \cap B) = P(B \cap A) \tag{4}$$

$$P(A \cap B) = P(A \mid B)P(B) \tag{5}$$

$$P(B \cap A) = P(B \mid A)P(A) \tag{6}$$

$$P(A \mid B)P(B) = P(B \mid A)P(A) \tag{7}$$

Hence, Bayes' Rule is given by:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} \tag{8}$$

With Bayes' Rule established, we transition from the abstract events, A and B, to the concepts of experimentally observed events (Evidence, E) and proposed hypotheses (H). In this discrete setting, both observed evidence and hypotheses can be represented as events within a shared probability space. Hypotheses are characterized as events that are evaluated in light of the observed data.

$$P(H \mid E) = \frac{P(E \mid H)P(H)}{P(E)}, \text{ where } P(E) > 0 \tag{9}$$

Total Probability:

Above, $P(E)$ is the total probability of the evidence occurring, taking into account all possible hypotheses. Now, suppose $\{H_1, H_2, \ldots, H_n\}$ is a mutually exclusive and collectively exhaustive set of hypotheses, each with positive probability. Because these hypotheses form a partition of the sample space, the sum of their probabilities is 1. For each hypothesis, there will be some probability of the evidence in question occurring, given that hypothesis (a conditional probability). For clarity, here the evidence is conditioned on that hypothesis. We now have one conditional probability for each hypothesis: namely, the probability of the evidence, E, given that hypothesis. This can be expressed for n hypotheses as follows:

$$P(E) = P(E \mid H_1)P(H_1) + P(E \mid H_2)P(H_2) + ... + P(E \mid H_n)P(H_n) \tag{10}$$

Or more succinctly,

$$P(E) = \sum_{i=1}^{n} P(E \mid H_i)P(H_i) \tag{11}$$

With this established, Bayes' Rule can be expressed as follows for a given hypothesis, H_j:

$$P(H_j \mid E) = \frac{P(E \mid H_j)P(H_j)}{\sum_{i=1}^{n} P(E \mid H_i)P(H_i)}, \text{ where } j \in \{1, 2, \ldots, n\} \text{ and } P(E) > 0 \tag{12}$$

Bayes' Rule: A Geometric Interpretation:

The foregoing states that the probability of a hypothesis, given evidence is the conditional probability of the evidence given that particular hypothesis (H_j), multiplied by the prior probability of that hypothesis, over the total probability of the evidence. This can be visually represented as follows:

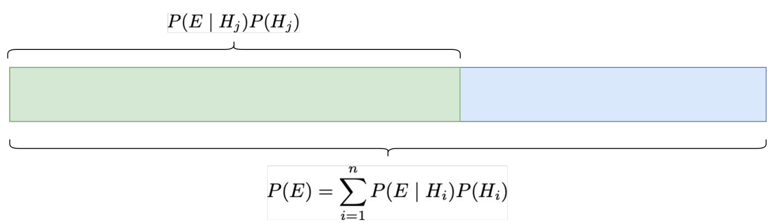

Figure 2: A visual representation of Bayes' Rule

Now, let's lay the ground work for a geometric interpretation. Bayes' Rule can be rearranged as follows:

Given:

$$P(E \mid H) = \frac{P(E \cap H)}{P(H)}, \text{ where } P(H) > 0, \tag{13}$$

Substituting this for all occurrences in equation 12, it can be shown that:

$$P(H_j \mid E) = \frac{P(E \cap H_j)}{\sum_{i=1}^{n} P(E \cap H_i)}, \text{ where } j \in \{1, 2, \ldots, n\} \text{ and } P(E) > 0 \tag{14}$$

This states that probability of hypothesis H_j, given evidence, E, is the probability of the intersection of the evidence with H_j over the sum of the probabilities of all intersections between E and the hypotheses. As a visual demonstration, take the case where there are 4 hypotheses.

Suppose a hypothesis-evidence space, in which the x-axis is partitioned into hypotheses, with each subsection's width representing $P(H_i)$, and the y-axis is a conditional-probability scale for evidence given a hypothesis. The boundary locations on the x-axis are therefore cumulative sums of the hypothesis probabilities.

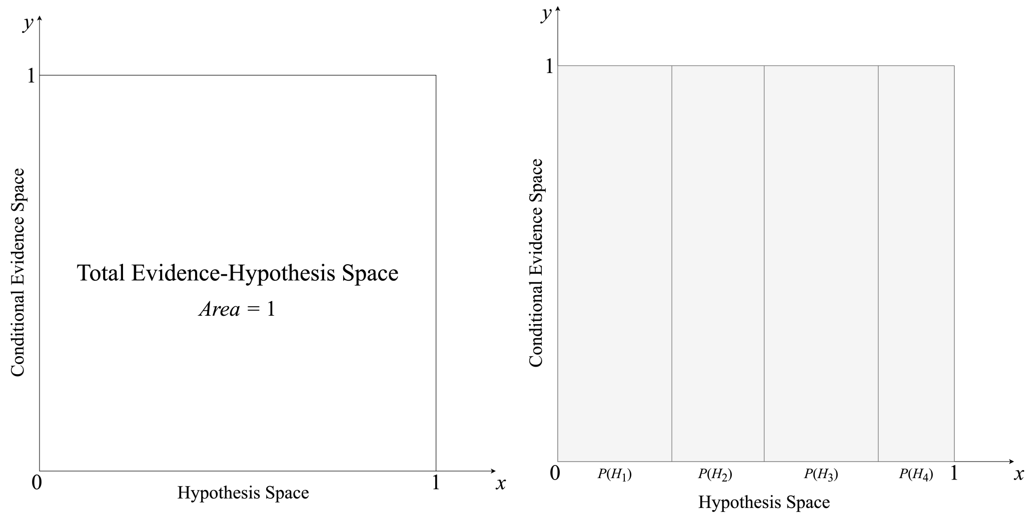

Figure 3: The hypothesis-evidence space divided by hypotheses.

Shown above, the hypothesis space can be divided into all relevant hypotheses, denoted by the subsections above.

Each hypothesis will have its own probability of evidence, previously referenced as $P(E \mid H_i)$ in the equation 11 above. These conditional probabilities can be represented as the y-axis component of the joint probability sections below.

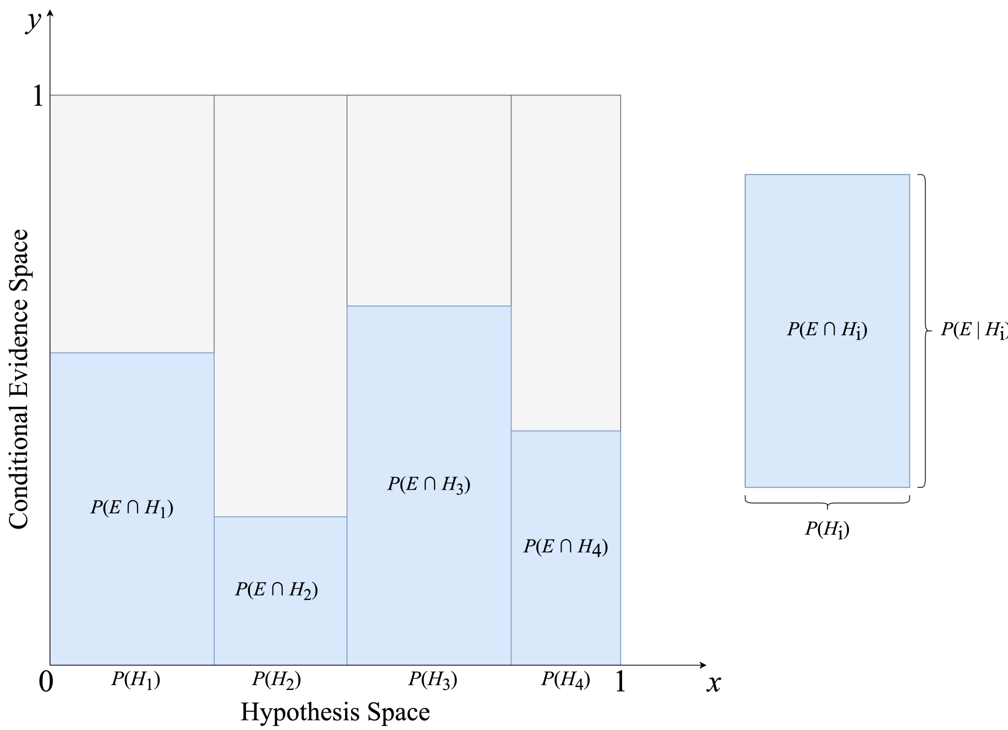

Figure 4: Hypothesis-evidence intersections visualized

Examining a single, arbitrarily-chosen subsection, note that the subsection's area is equal to the probability of the hypothesis multiplied by the probability of evidence, given that specific hypothesis. This finding is consistent with equation 5 above.

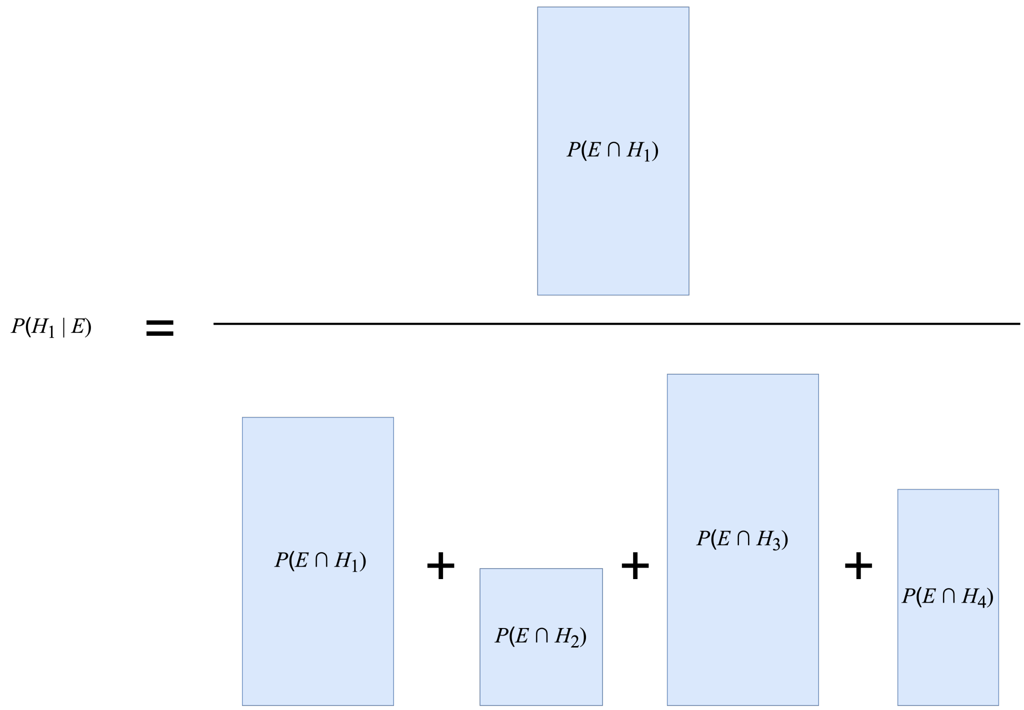

Figure 5: Probability of first hypothesis given evidence, visualized.

Bayes' Theorem with Continuous Random Variables:

The above treats probabilities associated with a finite collection of events. In many applications, however, the quantity of interest is an unknown continuous parameter. To mathematically express uncertainty about such a parameter, we treat the parameter as a random variable and describe it with a probability distribution. Using a continuous parameter θ, Bayes' Theorem for the posterior density is expressed as:

$$\pi(\theta \mid x) = \frac{f(x \mid \theta)\pi(\theta)}{\int_{\Theta} f(x \mid \theta)\pi(\theta)d\theta} \tag{15}$$

It's worth taking some time to understand what this equation is expressing. Intuitively, we know that Bayes' Theorem incorporates information from both experiment and prior beliefs, but to what end? We've heard that a bayesian update is the process by which a prior distribution is updated to a posterior distribution in light of experimental evidence, but how exactly is that achieved? In an effort to gain a deeper understanding, let's first understand the numerator of the theorem, as the denominator is simply the integral of the numerator with respect to θ.

Likelihood

The term, $f(x \mid \theta)$, viewed as a function of θ for fixed observed data x, is referred to as the likelihood. Here, the likelihood function evaluates how well different parameter values explain the observed data under a statistical model. It assigns relative support to parameter values; it does not, by itself, give probabilities of those parameter values. This definition involves some 'statistical model', against which the observed data can be assessed.

Model Selection Let's piece these concepts together with an example. Suppose that we're modeling the 'time to an event', say, the operating lifetime of a certain type of component in a machine. Also, suppose that we have reason to believe the lifetime of that component can be modeled as an exponential distribution:

$$f(x \mid \theta) = \theta e^{-\theta x}, \text{ where } \theta > 0 \text{ and } x \geq 0 \tag{16}$$

Here, $f(x \mid \theta)$ is a function of x and θ, where θ is the rate parameter. The greater the value of θ, the faster the function decays toward 0. At this point, we've proposed a statistical model as an attempt to explain the lifetime of the component type of interest. Let's call this proposal *model selection*. With a selected model, we now have something to compare experimental data against.

Experiment Suppose that an experiment is conducted to measure the lifetimes of the component type of interest. Intuitively, the empirical data outputted from such an experiment should certainly be used to inform the model we've proposed. Such data can be represented as a sample of lifetimes, $x_1, x_2, ..., x_n$.

Calculating Likelihood Using the set of empirically obtained lifetimes, we can now evaluate each observed lifetime x_i against our model, which can be expressed as:

$$f(x_i \mid \theta) = \theta e^{-\theta x_i} \tag{17}$$

This expression can be viewed as the likelihood function for a given x_i. Given that x_i is known (it came from the experiment), $f(x_i \mid \theta)$ condenses down to a function of θ. The statistical model we're evaluating against is indeed a probability density function over x values (component lifetimes) when θ is a fixed, constant value. That said, equation 17 above can be read in a second way: it fixes a known x_i value and allows θ to vary, resulting in $f(x_i \mid \theta)$ as solely a function of θ. It is this act of providing known x_i values that produces a likelihood function. A likelihood function is not, in general, a probability density function over θ, and it need not integrate to 1 with respect to θ. After evaluating each x_i against our model, we have a set of likelihood functions:

$$\{f(x_1 \mid \theta), f(x_2 \mid \theta), ..., f(x_n \mid \theta)\} = \{\theta e^{-\theta x_1}, \theta e^{-\theta x_2}, ..., \theta e^{-\theta x_n}\} \tag{18}$$

Assume each observation x_i is conditionally independent of the others given θ, and that all observations are drawn from the same exponential model. Therefore, given θ, the joint probability density of observing all lifetimes can be represented as:

$$f(x_1, \ldots, x_n|\theta) = \prod_{i=1}^{n} f(x_i|\theta) \tag{19}$$

Or in our example,

$$f(x_1, \ldots, x_n|\theta) = \theta^n e^{-\theta \sum_{i=1}^{n} x_i} \tag{20}$$

Now, with observed data as given and allowing theta to vary, the same expression can be interpreted through the lens of a likelihood function:

$$L(\theta|x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i|\theta) = \theta^n e^{-\theta \sum_{i=1}^{n} x_i} \tag{21}$$

In summary, we've arrived at an expression, in terms of θ, informed by the experimental data set of lifetimes.

The Prior

Thus far, we've selected a model, gathered observed data from an experiment, and constructed a likelihood function in terms of the parameter of interest, θ. That said, prior to an experiment, we do not know the value of θ. To account for such uncertainty, we can treat it as a random variable and assign a probability distribution to its possible values. For example, because the rate parameter θ must be positive, we may represent θ with a gamma prior density $\pi(\theta)$, with shape $\alpha > 0$ and rate $\beta > 0$. Such a density is called the *prior*, as it reflects beliefs about the parameter of interest prior to gathering empirical data. Its analogue in cases where probabilities are known is $P(H)$. Similar to the likelihood function in the numerator of Bayes' Theorem, the prior density is also a function of θ.

A Geometric Interpretation

We've demonstrated above that the numerator of Bayes' Theorem consists of two functions of θ multiplied by each other. Deviating from the specifics of the above example, the following visually demonstrates the multiplication of a hypothetical likelihood function and prior density:

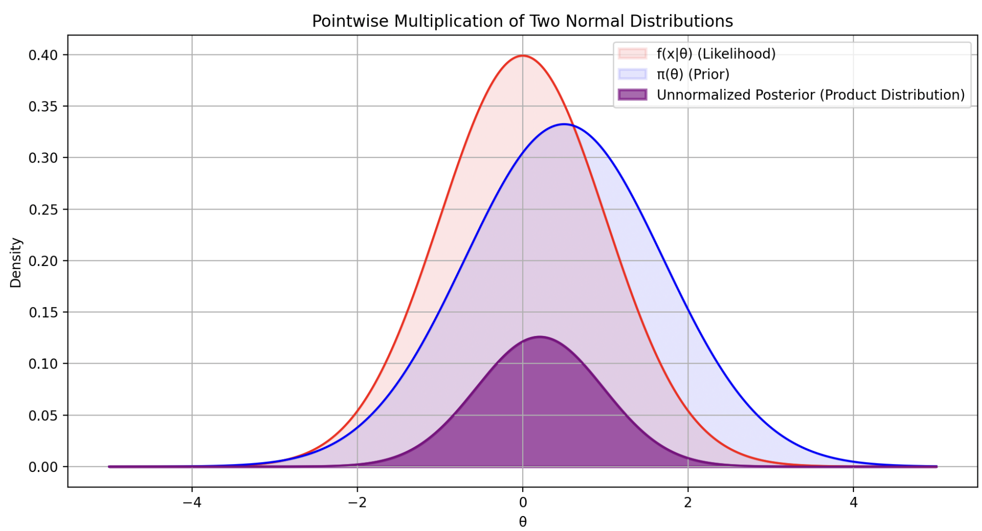

Figure 6: A hypothetical likelihood function, prior density, and their point-wise product

It's worth taking a moment to understand what the representation above expresses. In this particular example, the point-wise product appears to be centered between the hypothetical likelihood function and prior density. This positioning is not a general property of point-wise multiplication. The general property is that the product is large where both functions are large, thus accounting for information from both experiment (likelihood) and prior beliefs. The two functions are multiplied together by an operation called a point-wise multiplication. A point-wise multiplication of two functions involves multiplying their values at each point in their domain, resulting in a new function that reflects the combined characteristics of both original functions at each point. This process is visually demonstrated for a discrete case below:

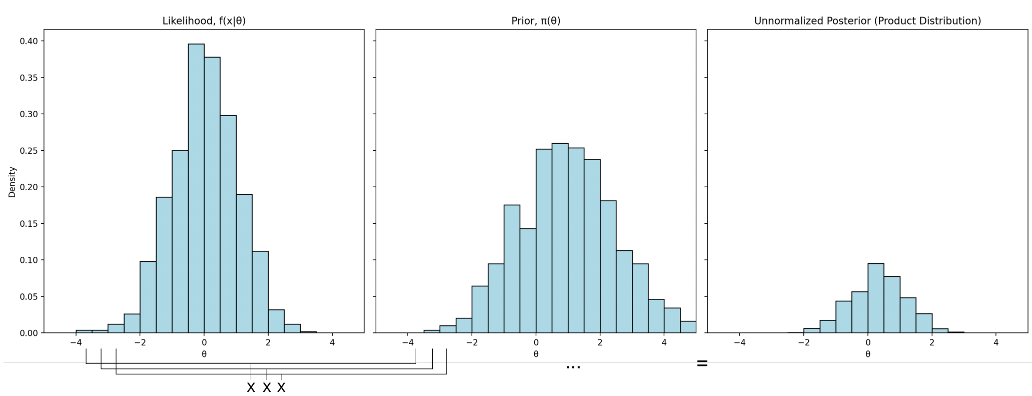

Figure 7: Point-wise multiplication visualized for a discrete case

The Posterior

While the point-wise product clearly incorporates data from both prior beliefs and experiment, it generally does not integrate to 1. At this moment, the product function is proportional to what will be the posterior density $\pi(\theta \mid x)$, but it need not integrate to 1. For that reason, it's called the *unnormalized posterior*. To obtain a valid posterior density, we need a function that achieves the following two properties: first, it must be proportional to the unnormalized posterior and second, it must integrate to 1. To achieve these properties, the unnormalized posterior can be divided by what's called the *normalizing constant*, which is the scalar value, $\int_\Theta f(x \mid \theta)\pi(\theta)d\theta$.

The Normalizing Constant

The following offers an algebraic explanation for why normalization requires dividing by the integral of the unnormalized posterior distribution function. Suppose we're given a nonnegative function, $g(x)$, such that $0 < \int_{-\infty}^{\infty} g(x)\,dx < \infty$, and we'd like to normalize it to $f(x)$. Normalization requires that $f(x)$ satisfies the following:

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \tag{22}$$

$$f(x) \propto g(x) \implies f(x) = kg(x), \text{ for some } k > 0 \tag{23}$$

Here, k is the unknown normalization factor. It can be found by substituting $kg(x)$ for $f(x)$ in equation 22,

$$\int_{-\infty}^{\infty} kg(x)\,dx = 1 \tag{24}$$

$$k\int_{-\infty}^{\infty} g(x)\,dx = 1 \tag{25}$$

$$\int_{-\infty}^{\infty} g(x)\,dx = \frac{1}{k} \tag{26}$$

$$k = \frac{1}{\int_{-\infty}^{\infty} g(x)\,dx} \tag{27}$$

$$\implies f(x) = \frac{g(x)}{\int_{-\infty}^{\infty} g(x)\,dx} \tag{28}$$

As expected, integrating $f(x)$ yields 1, satisfying equation 22.

$$\int_{-\infty}^{\infty} f(x)\,dx = \frac{\int_{-\infty}^{\infty} g(x)\,dx}{\int_{-\infty}^{\infty} g(x)\,dx} = 1 \tag{29}$$

This concludes the normalization of $g(x)$. In practice, the normalizing constant may be difficult to compute, which gives rise to probabilistic sampling methods, such as the Metropolis-Hastings and Gibbs Sampling algorithms.

Conclusion

It is my hope that this document serves as a useful tool for better understanding some of the mechanics involved in Bayes' Theorem. As a fan of open-source software, the Latex for this document will be available on GitHub. If anything is incorrect or can be better explained, I strongly urge you to open a pull request! Thank you.